



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

Project Title: AI-Based Smart Quiz System.

Course Code: CSE 4204

Course Title: Mobile Computing Lab

Submitted By:

Team Name : CSE4204-8A-T03

Section : 8A

GitHub Repository : <https://github.com/sudipto224/AI-based-Smart-Quiz-System>

Submitted to:

Md. Riaz Mahmud

Designation: Assistant Professor

Department of CSE

Date of Submission: 06/07/2026

TEAM INFORMATION

Official Team Name	CSE4204-8A-T03
Section	8A
Project Title	AI-Based Smart Quiz System
GitHub Repository	https://github.com/sudipto224/AI-based-Smart-Quiz-System

Team Information:

Role	Full Name	Student ID
Team Leader + Backend	Sudipto Das	11220320830
Frontend	Chaon Das	11210320629
Artificial Intelligence	Rakibur Rahman Mishan	11220320829
Database	Aditi Roy	11220320846

Frontend Technology Stack

The frontend of the AI-Based Smart Quiz System has been developed using a modern JavaScript ecosystem integrated with the Laravel backend through Inertia.js. The table below summarises the key technologies and tools used in the frontend development process.

Component	Technology Used
Framework	React.js 18.x (via Inertia.js)
Build Tool	Vite 5.x
Styling	Tailwind CSS 3.x
Routing	Inertia.js + Laravel Routes
State Management	React Hooks + Inertia useForm
HTTP Client	Axios (via Inertia)
Authentication	Laravel Breeze (session-based)
Fonts	Google Fonts (Figtree)

Completed Pages

All frontend pages required for the current phase of the project have been fully designed, developed, and integrated with the backend. The completed pages, along with their respective routes, are listed below.

No.	Page Name	Route
1	Landing/Home Page	/

No.	Page Name	Route
2	Login Page	/login
3	Registration Page	/register
4	Teacher Dashboard	/dashboard
5	Student Dashboard	/dashboard
6	Profile Page	/profile
7	Quiz Page	/quiz/{course_id}
8	Result Page	/result/{attempt_id}
9	AI Question Generation Page	/teacher/courses/{course_id}/questions/create
10	Suspicious Attempts Page	/teacher/suspicious-attempts
11	Course Management	/teacher/courses
12	Question Management	/teacher/courses/{course_id}/questions

Backend APIs Integrated

The frontend has been successfully connected to the backend through the following API endpoints. All listed endpoints have been tested and are functioning as expected.

Category	Endpoints
Authentication	POST /register, POST /login, POST /logout
Dashboard & Profile	GET /dashboard, GET /profile, POST

Category	Endpoints
	/profile/update
Course Management	GET /teacher/courses, POST /teacher/courses, PUT /teacher/courses/{id}, DELETE /teacher/courses/{id}
Question Management	GET /teacher/courses/{course_id}/questions, POST /teacher/courses/{course_id}/questions, PUT /teacher/questions/{id}, DELETE /teacher/questions/{id}
AI Question Generation	POST /teacher/generate-ai-question (Groq API)
Suspicious Attempts	GET /teacher/suspicious-attempts
Quiz	GET /quiz/{course_id}, POST /quiz/submit
Results & Leaderboard	GET /result/{attempt_id}, GET /leaderboard/{course_id}

Authentication Flow

The authentication system has been implemented using Laravel Breeze with session-based authentication, integrated seamlessly with the Inertia.js frontend. The complete authentication flow consists of the following five steps.

1. User Registration: The user fills in their name, email, and password, and may optionally check the “Register as Teacher” option. A POST request is sent to /register. Upon successful registration, the user is automatically logged in and redirected to the dashboard.

2. User Login: The user enters their registered email and password. A POST request is sent to /login. A successful login redirects the user to the dashboard according to their role (is_teacher).
3. Protected Routes: All authenticated routes are protected by the auth middleware, and teacher-specific routes are additionally protected by the teacher middleware. Unauthenticated users attempting to access protected routes are redirected to the login page.
4. Logout: A POST request is sent to /logout, which destroys the active session and redirects the user back to the login page.
5. Role-Based Redirection: Teachers are directed to the Teacher/Dashboard, which provides course management and suspicious attempts monitoring, while students are directed to the Student/Dashboard, which displays available courses.

Current Development Progress

As of the submission date, all planned frontend development tasks for this phase have been completed. The table below outlines the progress of each module.

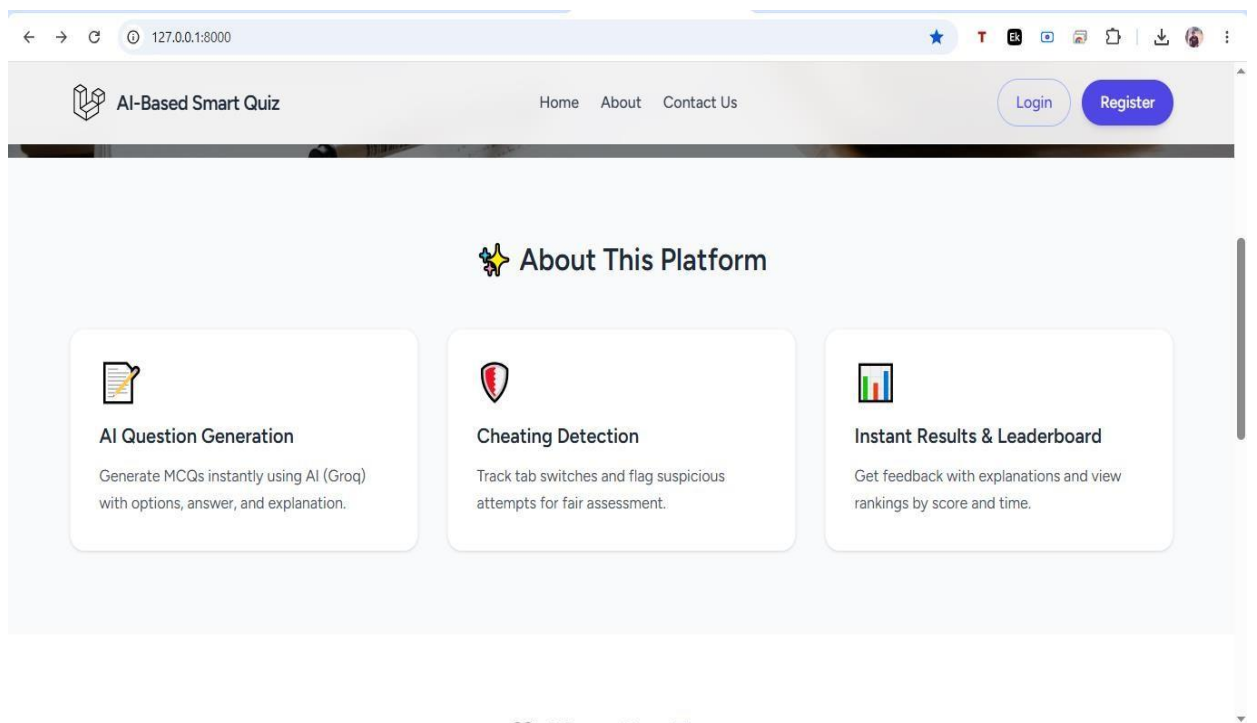
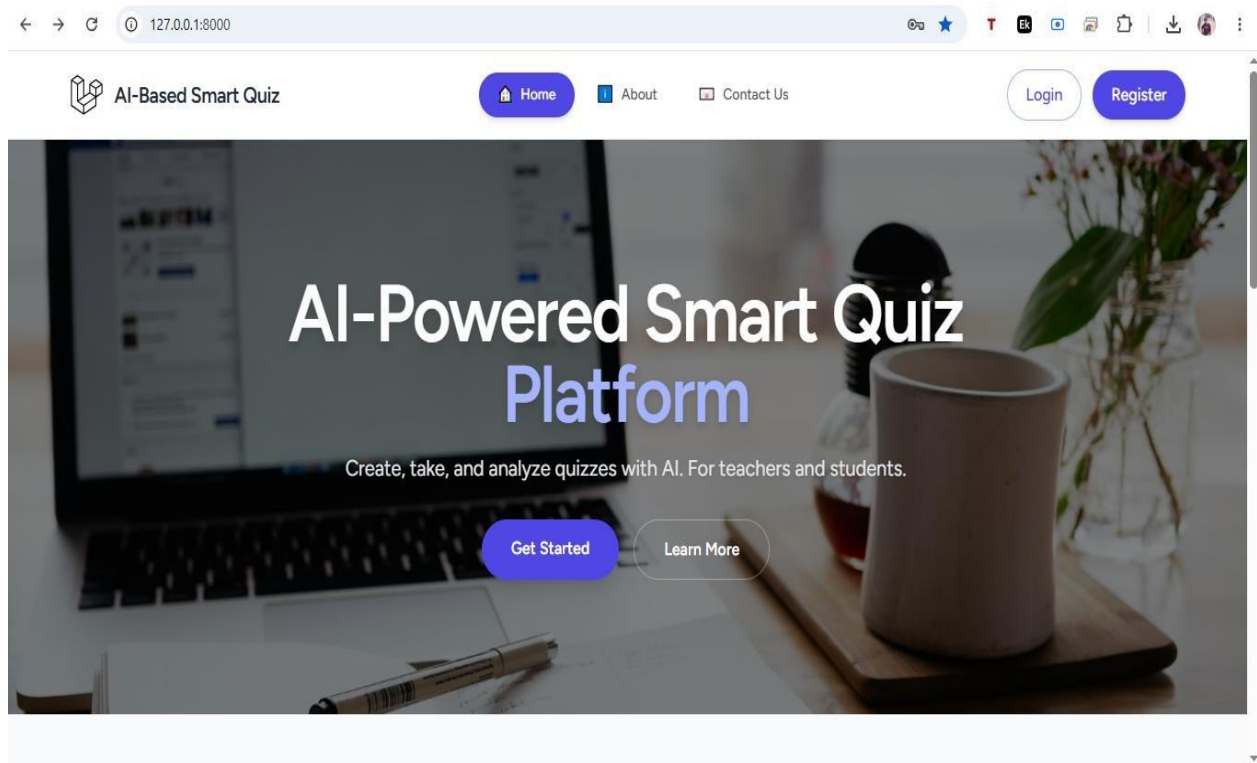
No.	Module/Task	Status
1	Frontend Project Setup (React + Inertia)	100% Complete
2	Landing/Home Page	100% Complete
3	Login Page	100% Complete
4	Registration Page	100% Complete
5	Teacher Dashboard	100% Complete
6	Student Dashboard	100% Complete
7	Profile Page	100% Complete
8	Quiz Page (with Timer)	100% Complete

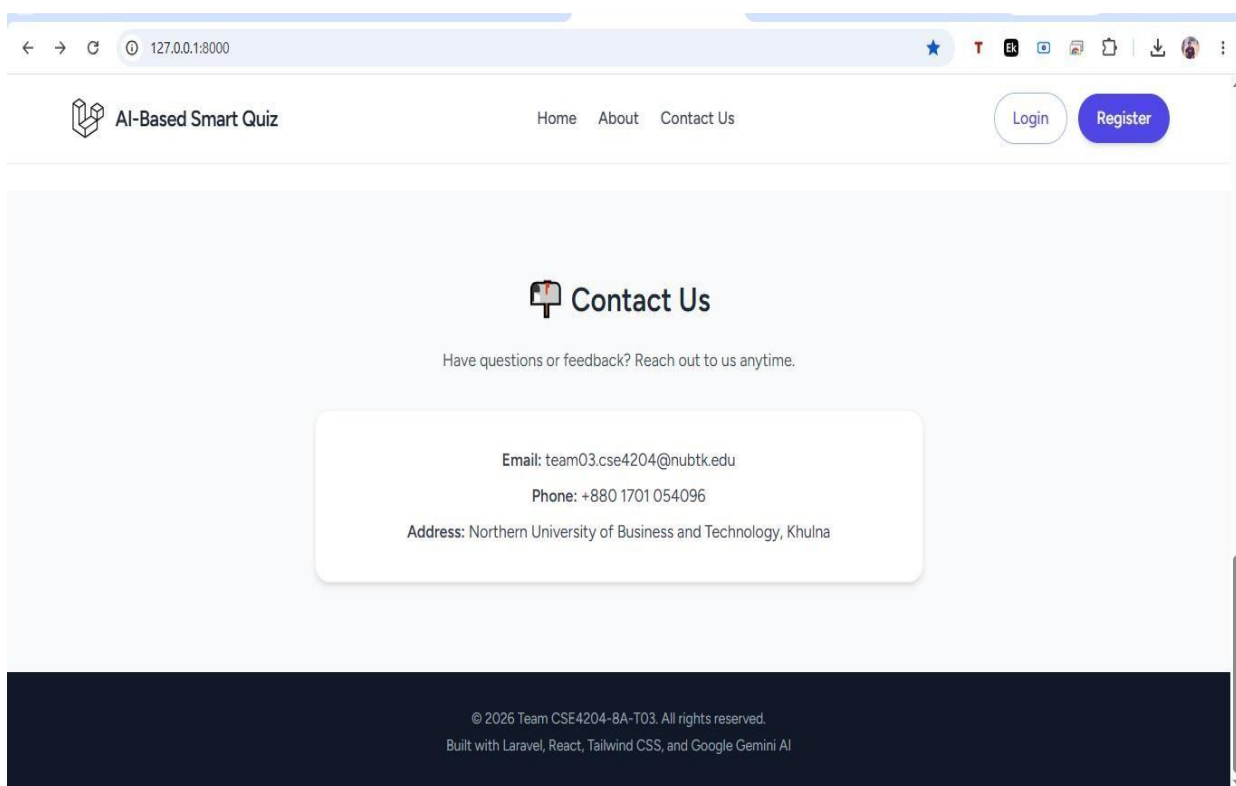
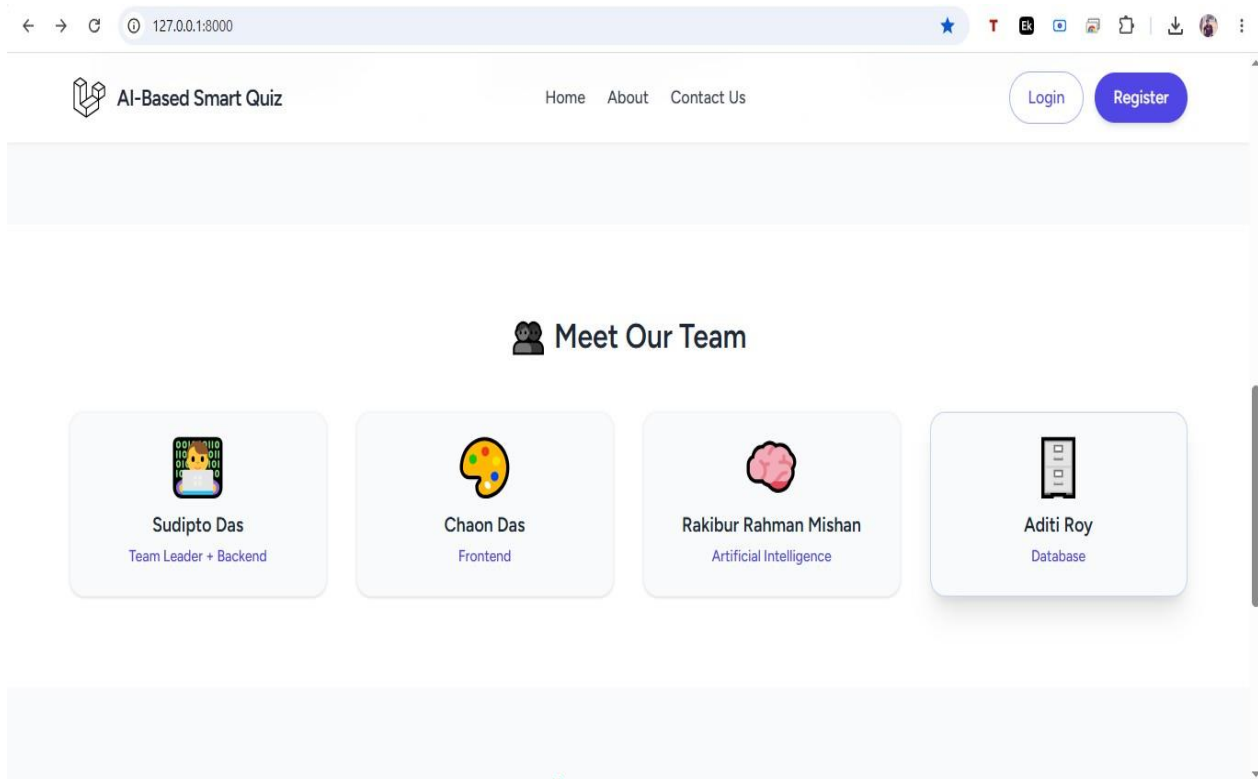
No.	Module/Task	Status
9	Result Page	100% Complete
10	AI Question Generation (Groq API)	100% Complete
11	Suspicious Attempts Page	100% Complete
12	Course CRUD	100% Complete
13	Question CRUD	100% Complete
14	Backend API Integration	100% Complete
15	Authentication Flow	100% Complete
16	Form Validation & Error Messages	100% Complete
17	Loading Indicators	100% Complete
18	Success/Error Toast Messages	100% Complete
19	Responsive Design (Desktop, Tablet, Mobile)	100% Complete
20	GitHub Repository Maintenance	95% Complete

Screenshots

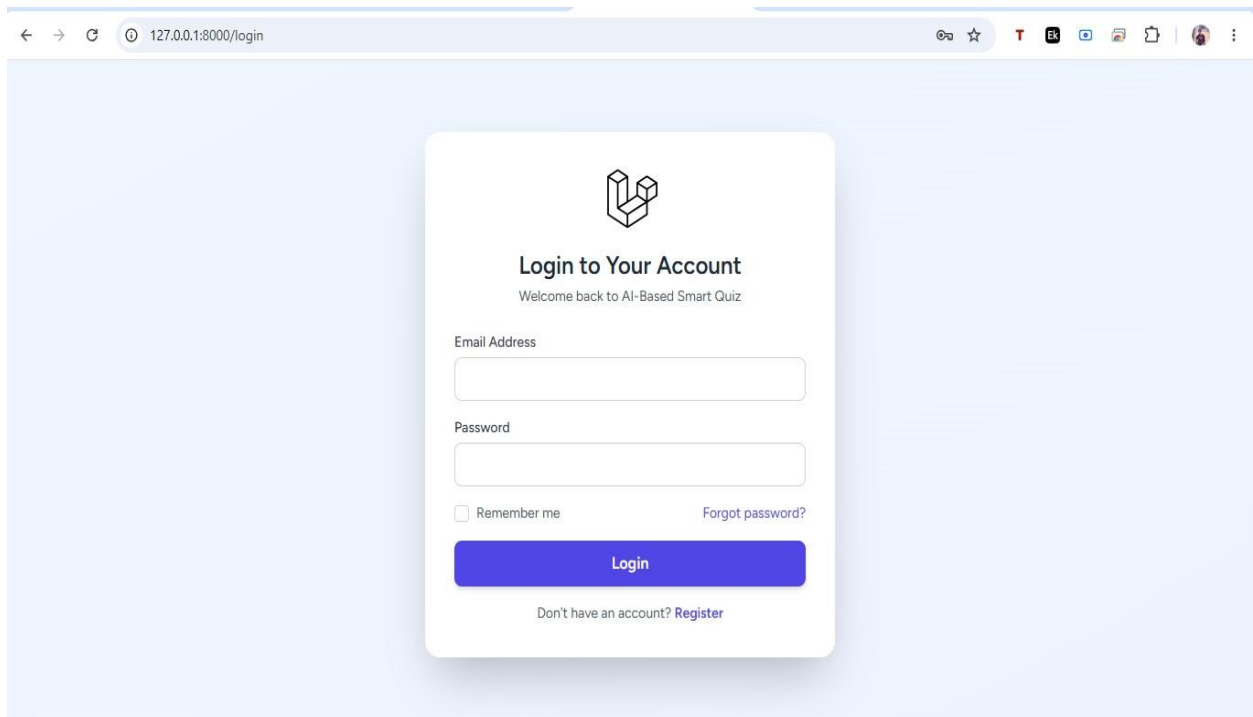
The following section provides placeholders for screenshots of each completed page. Screenshots are to be inserted manually below the corresponding label.

1. Landing/Home Page (Desktop)





2. Login Page



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/login". The page features a light blue background with a white login form in the center. The form has a logo at the top, followed by the title "Login to Your Account" and a subtitle "Welcome back to AI-Based Smart Quiz". Below this are input fields for "Email Address" and "Password". There is a checkbox for "Remember me" and a link for "Forgot password?". A blue "Login" button is at the bottom of the form, with a link "Don't have an account? Register" below it.

127.0.0.1:8000/login



Login to Your Account

Welcome back to AI-Based Smart Quiz

Email Address

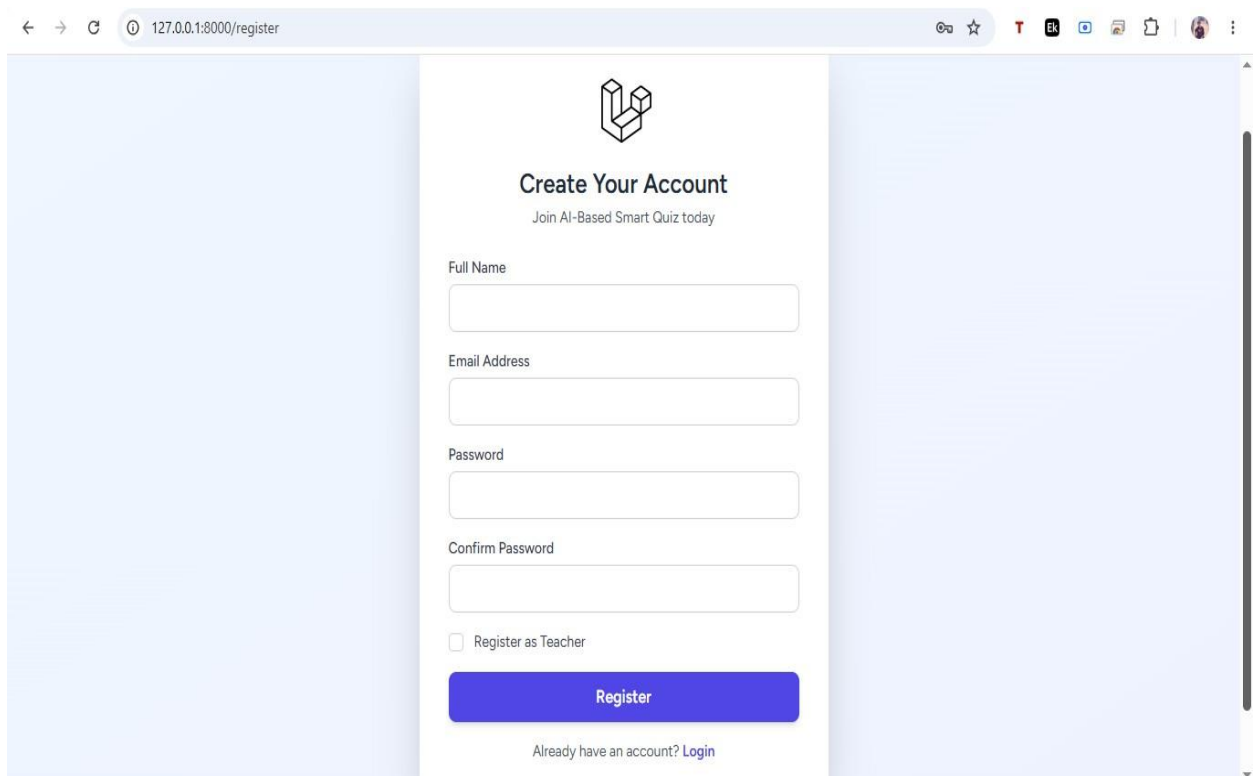
Password

☐ Remember me [Forgot password?](#)

Login

Don't have an account? [Register](#)

3. Registration Page



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/register". The page features a light blue background with a white registration form in the center. The form has a logo at the top, followed by the title "Create Your Account" and a subtitle "Join AI-Based Smart Quiz today". Below this are input fields for "Full Name", "Email Address", "Password", and "Confirm Password". There is a checkbox for "Register as Teacher" and a blue "Register" button. A link "Already have an account? Login" is at the bottom of the form.

127.0.0.1:8000/register



Create Your Account

Join AI-Based Smart Quiz today

Full Name

Email Address

Password

Confirm Password

☐ Register as Teacher

Register

Already have an account? [Login](#)

4. Teacher Dashboard

The screenshot shows a web browser at the address 127.0.0.1:8000/dashboard. The page header includes the site name 'AI-Based Smart Quiz' and navigation links for 'Dashboard', 'Profile', and 'Logout'. The main content area features a 'Welcome, Teacher!' message with a hand icon. Below this are two prominent buttons: a blue 'Manage Courses' button with a book icon and a red 'Suspicious Attempts' button with a person icon. A section titled 'Your Courses' contains a single entry, 'Mobile Computing (3 questions)', enclosed in a light gray box.

5. Student Dashboard

The screenshot shows a web browser at the address 127.0.0.1:8000/dashboard. The page header is identical to the teacher dashboard, with the site name 'AI-Based Smart Quiz' and navigation links for 'Dashboard', 'Profile', and 'Logout'. The main content area features a 'Welcome, Student!' message with a hand icon. Below this, a light gray box displays the course details: 'Mobile Computing', 'Mobile Computing quesations' (note the typo), and '25 sec/question'. A blue 'Start Quiz' button is positioned at the bottom of this box.

6. Profile Page

Student profile page –

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/profile". The page header includes the logo "AI-Based Smart Quiz" and navigation links for "Dashboard", "Profile", and "Logout". The main content area features a "My Profile" form with a user icon. The form includes a role dropdown set to "Student", a "Full Name" input field containing "test", and an "Email Address" input field containing "test@gmail.com". At the bottom of the form are "Update Profile" and "Cancel" buttons.

127.0.0.1:8000/profile

AI-Based Smart Quiz

Dashboard Profile Logout

My Profile

Role: Student

Full Name

test

Email Address

test@gmail.com

Update Profile Cancel

Teacher profile page –

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8000/profile". The page header includes the logo "AI-Based Smart Quiz" and navigation links for "Dashboard", "Profile", and "Logout". The main content area features a "My Profile" form with a user icon. The form includes a role dropdown set to "Teacher", a "Full Name" input field containing "Teacher", and an "Email Address" input field containing "teacher5@gmail.com". At the bottom of the form are "Update Profile" and "Cancel" buttons.

127.0.0.1:8000/profile

AI-Based Smart Quiz

Dashboard Profile Logout

My Profile

Role: Teacher

Full Name

Teacher

Email Address

teacher5@gmail.com

Update Profile Cancel

7. Quiz Page

← → ↻ 127.0.0.1:8000/quiz/2 🔍 ☆ T Ek 📄 📁 📌 👤 ⋮

AI-Based Smart Quiz Dashboard Profile Logout

Mobile Computing

Question 1 of 3

⌚ 16 sec

What is the primary purpose of using a heartbeat protocol in a mobile ad-hoc network?

☐ A. To maintain network connectivity through periodic 'hello' messages.

☐ B. To prioritize data packets based on their size and transmission speed.

☐ C. To compress data packets to reduce transmission overhead.

☐ D. To encrypt data packets for enhanced security.

Next →

8. Result Page

← → ↻ 127.0.0.1:8000/result/1 🔍 ☆ T Ek 📄 📁 📌 👤 ⋮

AI-Based Smart Quiz Dashboard Profile Logout

🎯 Your Score

0 / 3

Time taken: 00:00:36

0

📄 Detailed Review

1. What is the primary purpose of using a heartbeat protocol in a mobile ad-hoc network?

Your answer: (Not answered)

Correct answer: A. To maintain network connectivity through periodic 'hello' messages.

In mobile ad-hoc networks, the heartbeat protocol is used to maintain network connectivity by sending periodic 'hello' messages. This helps in detecting node failures and maintaining the network topology.

2. What is the primary purpose of using a sandbox environment in mobile computing?

Your answer: (Not answered)

Correct answer: A. To enhance the security of mobile applications by isolating them from the underlying system

9. AI Question Generation Page

127.0.0.1:8000/teacher/courses/2/questions/create

+ Add Question – Mobile Computing

AI Question Generator

Enter topic (e.g., Mobile IP) **Generate with AI**

Generate a complete MCQ with options, correct answer, and explanation.

Question

Option 1

Option 2

Option 3

Option 4

Correct Answer (exact text)

Explanation

127.0.0.1:8000/teacher/courses/2/questions/create

+ Add Question – Mobile Computing

AI Question Generator

Mobile Computing **Generate with AI**

Generate a complete MCQ with options, correct answer, and explanation.

Question

What is the primary purpose of using a Virtual Private Network (VPN) in mobile computing?

Option 1

A. To enhance the mobile device

Option 2

B. To improve the security and p

Option 3

C. To increase the mobile device

Option 4

D. To optimize the mobile device

Correct Answer (exact text)

B. To improve the security and privacy of data transmission over public

Explanation

A VPN is used to create a secure and encrypted connection between a mobile device and a remote network, protecting data from

10. Suspicious Attempts Page

← → ↻ 127.0.0.1:8000/teacher/suspicious-attempts 🔍 ☆ T Ek 📄 📁 📧 | 👤 ⋮

AI-Based Smart Quiz Dashboard Profile Logout

Suspicious Attempts

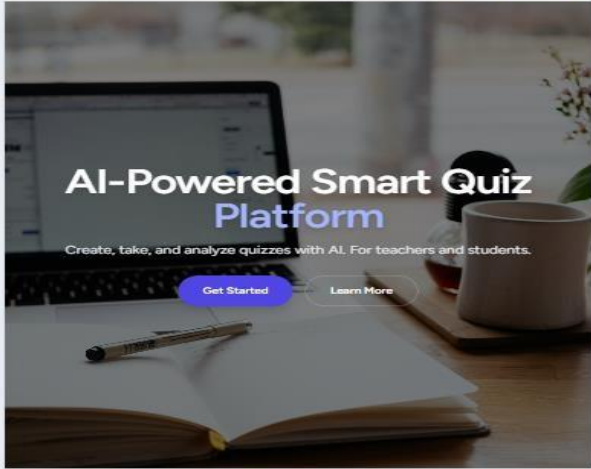
Student	Course	Score	Tab Switches	Avg Time	Attempted At
test	Mobile Computing	0/3	0	2.1 sec	7/6/2026, 1:01:41 AM
test	Mobile Computing	0/3	0	3.2 sec	7/5/2026, 11:56:15 PM
test	Mobile Computing	1/3	0	1.4 sec	7/5/2026, 11:51:40 PM

11. Mobile Responsive View (Landing Page)

← → ↻ 127.0.0.1:8000

Dimensions: Responsive ▾ 768 × 1024 50% 📐 No throttling ▾ 'Save-Data': default 🔄

AI-Based Smart Quiz Home About Contact Us Login Register



AI-Powered Smart Quiz Platform

Create, take, and analyze quizzes with AI. For teachers and students.

Get Started Learn More

 About This Platform

GitHub Repository Link

The complete source code for the AI-Based Smart Quiz System, including all frontend and backend commits, is maintained and version-controlled on GitHub at the following repository link:

<https://github.com/sudipto224/AI-based-Smart-Quiz-System>